Visual Analysis of Cardiac Arrest Prediction Using Machine Learning Algorithms: A Health Education Awareness Initiative

ABSTRACT

A visual analysis may accurately predict cardiac arrest, making it a potent educational tool for raising public awareness of health issues. By predicting cardiac arrest earlier, preventative steps can be taken to save lives and the dissemination of such health knowledge can dramatically lower the world mortality rate. A heart attack, also known as cardiac arrest, encompasses various heart-related disorders and has been the leading cause of death worldwide in recent decades. Several medical data mining and machine learning technologies are being applied to gather helpful knowledge regarding heart disease prediction. The accuracy of the intended outcomes, however, is insufficient. This chapter aims to predict the likelihood of patients having heart disease to solve the issue. Specifically, it compared alternative models for the identification of cardiac arrest to appropriately categorize and forecast heart attack instances with compact features. The use of ensemble algorithms over classifier algorithms gives a maximum accuracy of 96.5%, which is examined in our investigation.

Keywords: Cardiac Arrest, Machine Learning, Comparative Analysis, Heart Disease Prediction, Data Standardization, Supervised Learning, Health Education Awareness

INTRODUCTION

Visual analysis is an effective technique for educating society. Raising public awareness on various health issues can lead to taking preventative actions. For instance, through earlier cardiac arrest prediction, curative measures can be taken in advance to save human life (Maaliw, Alon, Lagman, Garcia, Susa, et al., 2022). Promoting such health knowledge in society can dramatically lower the world mortality rate. According to the World Health Organization (2021), cardiovascular diseases (CVDs) are the leading cause of mortality globally, killing an estimated 17.9 million people each year. Heart attacks and strokes are responsible for 85% of these fatalities. CVDs include coronary heart disease, cerebrovascular illness, rheumatic heart disease, and other diseases (Amini et al., 2021). Nearly three-quarters of all heart-related fatalities occur in low- and middle-income nations. In 2015, low- and middle-income nations accounted for 82% of the 17 million early deaths (before the age of 70) related to non-communicable illnesses, accounting for a total of 37%. Four out of every five CVD fatalities are caused by heart attacks or strokes, among them, one-third of the deaths occur in those below 70-year old patients (Desai et al., 2021). High blood pressure, cholesterol, and lipid levels, as well as being overweight or obese, are all signs of heart illness (Jurgens et al., 2022). Identifying patients who are more likely to suffer from cardiac arrest and ensuring that they receive proper care would assist to prevent early deaths. All primary health care providers should give access to important noncommunicable disease drugs and basic health technologies to ensure that individuals in need receive treatment and counselling. Most heart attacks can be averted by addressing various behavioural risk factors such as cigarette use (Sandhu et al., 2012), poor nutrition and obesity (Garcia & Garcia, 2023), problematic alcohol consumption (Chudzińska et al., 2022), and lack of physical inactivity through population-wide initiatives.

Any serious cardiac ailment is referred to as cardiovascular disease. Because they can be fatal, researchers are concentrating on developing smart systems to precisely identify cardiac illnesses based on electronic health data, with the use of machine learning algorithms (Pal et al., 2022). For instance, Javeed et al. (2022) investigated several machine learning techniques for heart disease prediction that make use of patient data on key health indicators. Over the past few decades, cardiovascular illnesses have become the leading cause of mortality globally in both industrialized and developing nations (Bae et al., 2021; Ruan et al., 2018). The mortality rate can be decreased through early identification of heart disorders and ongoing clinical monitoring by professionals. However, because it takes more intelligence, effort, and knowledge, reliable identification of cardiac problems in all situations and 24-hour patient consultation by a doctor are not yet possible.

Thanks to the growing volume of data, machine learning is becoming more popular. Machine learning allows humans to gain information from vast amounts of data, which is difficult and often impractical for humans to do. Furthermore, and perhaps most importantly, intelligent optimization algorithms are used to compare the various machine learning techniques as stated in Figure 1. Throughout the world, machine learning is applied in numerous fields (Garcia et al., 2019; Maaliw, Susa, et al., 2022). No industry is more so than the medical field. Machine learning has the potential to be extremely useful in determining whether certain conditions, such as heart disease and locomotor disorders, will exist. A doctor may be able to make substantial adjustments to their diagnosis and course of therapy if such information is anticipated well in advance.

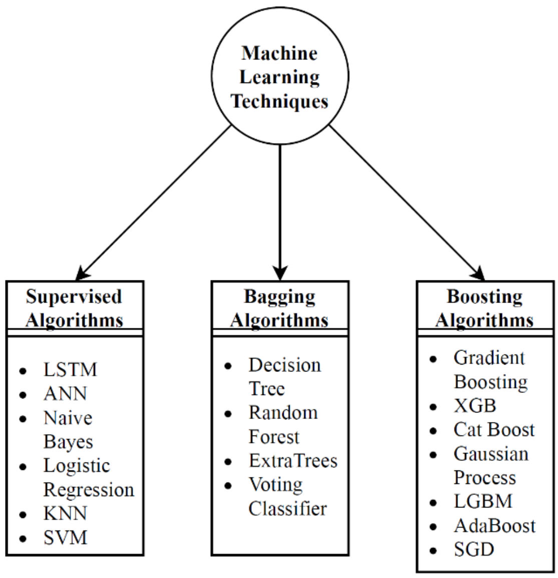


Figure 1. Broad Classification of Machine Learning Techniques

MAIN FOCUS OF THE CHAPTER

In this chapter, we did an exhaustive comparison and assessment of several classifiers for cardiac arrest prediction using three distinct baskets: bagging methods, boosting algorithms, and supervised algorithms. Worldwide, heart disease has been identified as a leading cause of death. Early disease detection is more crucial and valuable, and Auto-Prognosis significantly improved cardiac threat assessment performance when compared to well-performing systems. Emulation of the decision-making system that can identify cardiovascular illness using binary classification algorithms is part of the architecture of an intelligent system. An intelligent system attempts to identify the disease using human indications as inputs. Instead of using procedural code, a classic expert system relies on if-then-else logic to function. Instead of if-then-else rules, machine learning algorithms are utilized in the workplace to create intelligent systems. The remainder paper is structured with: Section 2 discussed earlier studies as well as state-of-art methods and strategies. Furthermore, pre-processing of data is discussed in Section 3, including dataset feasibility summary, feature visualizations, and attribute extraction. Section 4 digs into categorization models and the underlying principles of each method. Section 5 shows the efficiency measures, methods, and experimental setup. It also shows how the experiment was carried out and the results acquired. Section 6 finishes with a recap of previous work and many ideas for future improvements.

LITERATURE REVIEW

Simple lifestyle changes along with early care significantly increase the prognosis of heart attacks. The multi-factorial aspect of multiple contributory risk factors such as diabetes, high blood pressure, high cholesterol, and others makes it impossible to distinguish high-risk patients. Here is where data processing and deep learning come in handy. By analysing thousands of clinical reports and other medical data, machine learning algorithms can identify trends associated with diseases and health conditions (Banerjee et al., 2022; Maaliw, Alon, Lagman, Garcia, Abante, et al., 2022; Özbay Karakuş & Er, 2022).

Several studies using various classifiers and function selection methods are performed on medical data sets. For instance, in the computer science literature, various machine learning techniques have been applied to cardiac arrest prediction (Chae et al., 2022; Dissanayake & Md Johar, 2021). For the best analysis of cardiac disorders, Jindal et al. (2021) used a machine learning technique and 4% KNN (87% accuracy) is considerably more efficient than SVP, Decision tree, and linear regression algorithms. It also employs the usage of a confusion matrix to compare the accuracy of each method. Meanwhile, Junaid and Kumar (2020) suggested a hybrid algorithm-based prediction of heart disease in their research study. The sensitivity and specificity of this Hybrid are 91.47% and 82.11%, respectively. Their study provides a novel route for experiments by demonstrating how these principles might be used in smart devices and data mining to enhance cardiac arrest detection and treatment. Depending on the circumstance, different methods performed better, whether cross-validation, grid search, calibration, or feature selection were utilized or not. Any algorithm has the inherent ability to outperform other algorithms. The average accuracy of the highest efficiency without improving NB percentage is 83.6%, which is a high percentage of RF's 81.4% (Khourdifi & Bahaj, 2018). On the other hand, Karthikeyan et al. (2021) found that ANN has a higher percentage accuracy (85%) after comparing ANN, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine algorithms to diagnose heart diseases in patients. Once a much larger dataset is available, ANN may be employed. The number of hidden layers and epochs might also be raised, which would improve the neural network's throughput.

Another approach uses RNN and GRU to make the system more accurate and effective to predict silent heart attacks and inform the user at the earliest possible. This system increased the heart attack prediction accuracy to 92% and has proved to be an excellent source in predicting silent heart attacks (Kishore et al., 2018). The enhanced medical characteristics in the Dataset boost accuracy. Logistic

Regression, Random Forest Classifier, and KNN are the methods utilized to create the provided model. The accuracy was 87.5%, which is higher than the prior models' accuracy of 85%. As a result, as the number of qualities rises, so does the accuracy (Jindal et al., 2021). In this study, two additional input variables, obesity, and smoking produce more accurate outcomes. Decision trees, Naive Bayes, and Neural Networks were used as data mining classification approaches. According to the data, Neural Networks produce more accurate results than Decision Trees and Naive Bayes (Priya et al., 2022). Meanwhile, Sultana et al. (2016) addresses the issue of the prediction of heart disease according to some input attributes. Accuracy does not always give an accurate measure of the performance of a classifier, as FPR is a dangerous error. For this reason, the classifier performance is usually expressed using the Receiver Operating Characteristics (ROC) curve (that is, a plot of TPR vs FPR). The performance of the Bayes Net classifier is the best, and that of the J48 classifier is the worst. The performances of SMO, MLP, and KStar are also good. In another study, Rajendran and Karthi (2022) used three classifiers (ID3, CART, and DT) to diagnose patients with heart diseases. Observation shows that CART performance is having more accuracy when compared with the other two classification methods. The best algorithm based on the patient's data is CART Classification with an accuracy of 83.49%, and the total time taken to build the model is at 0.23 seconds. CART classifier has the lowest average error at 0.3 compared to others. These results suggest that the CART classifier can significantly improve the conventional classification methods used in the study.

In another study, Takci (2018) conducted experiments with and without feature selection to measure the feature selection effect for heart attack prediction. Without feature selection, the best result, based on model accuracy, gave many classifiers. Eight classifiers gave an accuracy of around 80%. BLR and naïve Bayes gave the best result in terms of processing time. The same algorithms also gave the best results according to model accuracy and processing time. Then the experiments were repeated by selecting features. With feature selection, even though not the case for all algorithms, some of them improved both processing time and model accuracy. The highest accuracy value was 82.59% without feature selection and it was improved to 84.81% with feature selection. SVM-linear and naïve Bayes gave model accuracy of 84.81%. On the other hand, the results in the field of data classification were obtained with the Naive Bayes algorithm, Decision list algorithm, and KNN algorithm, and overall, the performance made known the Naive Bayes Algorithm when tested on heart disease datasets (Rajkumar & Reena, 2010). Accordingly, it was found that the Naive Bayes algorithm was the best compact time for processing datasets and showed better performance in accuracy prediction. The time taken to run the data for the result is fast when compared to other algorithms. It shows the enhanced performance according to its attribute. Attributes are fully classified by this algorithm, and it gives 52.33% of accurate results.

In a study conducted by Chitra and Seenivasagam (2013), the sensitivity achieved for the FCM classifier is 91.53 with an average false positive of 0.9 per 30 records. The achieved accuracy is 92%, which is better than the performance of the neural network-based classifier and K-means clustering algorithm. The increase in the performance of the FCM clustering algorithm is because the weights of data attributes are set to adjust original samples to the uniform distribution, which could be suitable for the character of FCM calculation to improve the accuracy. The results of the classification experiment, performed over a data set obtained from 270 patients, show that the classifier has achieved better accuracy than most of the existing algorithms. Meanwhile, Methaila et al. (2014) focuses on using different algorithms and combinations of several target attributes for effective heart attack prediction using data mining. Decision Tree outperforms with 99.62% accuracy by using 15 attributes. Also, the accuracy of the Decision Tree and Bayesian Classification further improves after applying a genetic algorithm to reduce the actual data size to get the optimal subset of attributes sufficient for heart disease prediction. In principle, the BN learning algorithms can discover the mediated correlation since they test pairwise independence and conditional independence given values of other variables. Bayesian networks are a tool of choice for reasoning in uncertainty, with incomplete data. However, often, Bayesian network structural learning only deals with complete data. This study has proposed an adaptation of the learning process of the Chow–Liu, and TAN from incomplete and imbalanced datasets. These methods have been successfully tested on the dataset. It is seen that the TANI algorithm is a single winner with D.Bin (Salman, 2019). Finally, Dangare and Apte (2012) presented a Heart disease prediction system (HDPS) using data mining and artificial neural network (ANN) techniques. From the ANN, a multilayer perceptron neural network along with a Backpropagation algorithm was used to develop the system. The experimental result shows that by using neural networks the system predicts heart disease with nearly 100% accuracy.

Table 1. Sample Content from the Dataset Used for Validation

age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
44	1	1	120	263	0	1	173	0	0	2	0	3	1
52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

Note: The descriptions of the attributes are presented in Table 2.

Table 2. Description of the Attributes in the Dataset

Name	Type	Description
age	Continuous	Age in years
sex	Discrete	0 = female, 1 = male
cp	Discrete	Chest Pain Type: 1 = typical angina, 2 = atypical angina, 3 = non-angina pain, 4 = asymptom
trtbps	Continuous	Resting Blood Pressure (in mm Hg)
chol	Continuous	Serum Cholesterol (in mg/dl)
fbs	Discrete	Fasting Blood Sugar > 120mg/dl: 1 = True, 0 = False
restecg	Discrete	Rest Electrocardiograph: 0 = normal, 1 = having ST-T wave abnormality, 2 = left ventricular hypertrophy
thalachh	Continuous	Maximum Heart rate achieved
exng	Discrete	Exercised induced angina: 0 = no, 1 = yes

oldpeak	Continuous	Depression induced by exercise relative to rest
slp	Discrete	The slope of the peak exercise segment: 1 = up sloping, 2 = flat, 3 = down sloping
caa	Continuous	The number of Major Vessels coloured by fluoroscopy ranged between 0 and 3.
thal	Discrete	Thalassemia defect type: 3 = normal, 6 = fixed defect, 7 = reversible defect

SCENARIOS AND DATA DESCRIPTIONS

The Heart Disease dataset is utilized to conduct the validation of the proposed model. This Dataset was initially cited by the University of California, Irvine Repository (Mamun et al., 2022). This dataset comprises numerous attributes and a series of 303 entries collected from various heart patients' data as depicted in Table 1 and Table 2. The corresponding graphical visualizations are stated in Figure 2 and Figure 3. Based on the attribute selections, the quantitative measure among Attribute-to-Attribute Co-variance is depicted in Figure 4a and the covariance heatmap is visualized in Figure 4b.

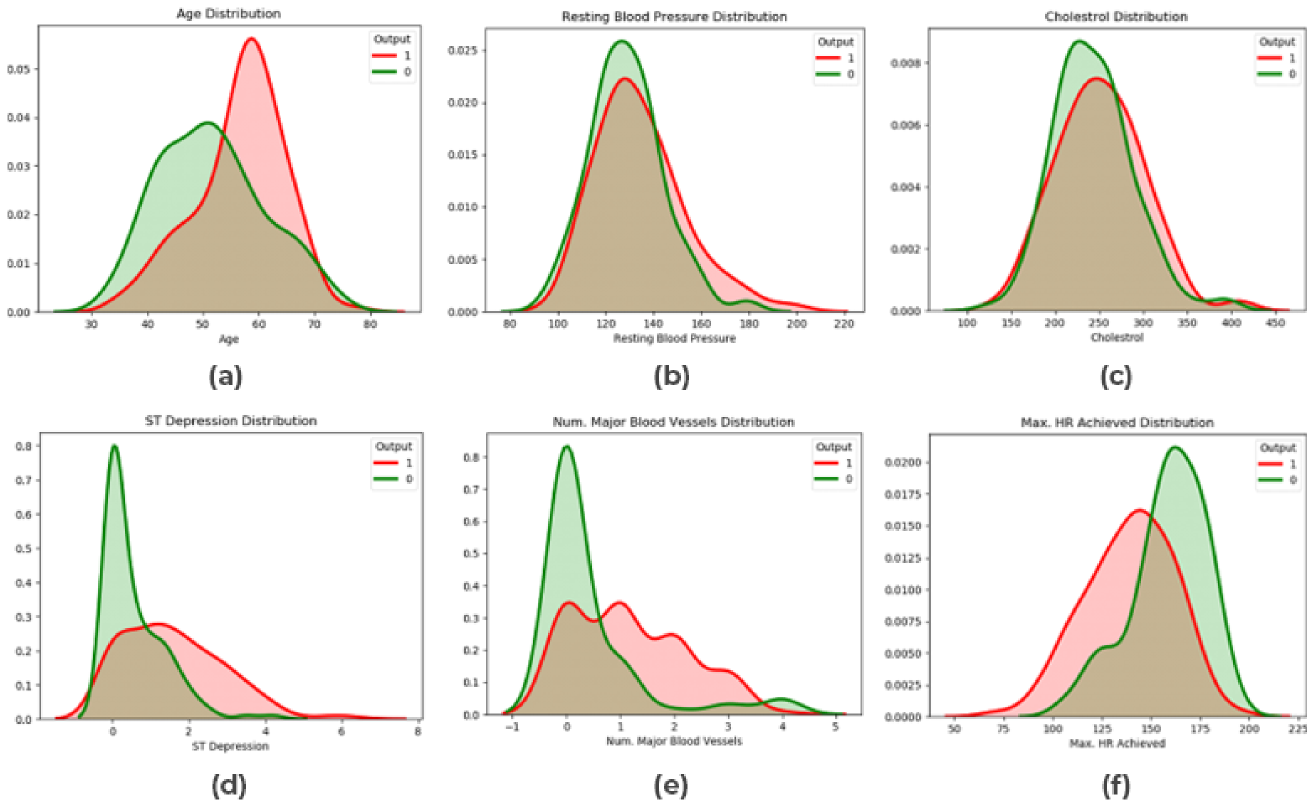


Figure 2. Distribution Plots for the Numeric Attributes of the Dataset: (a) Age, (b) Resting Blood Pressure, (c) Cholesterol, (d) ST Depression, (e) Major Blood Vessels, and (f) Max HR Achieved.

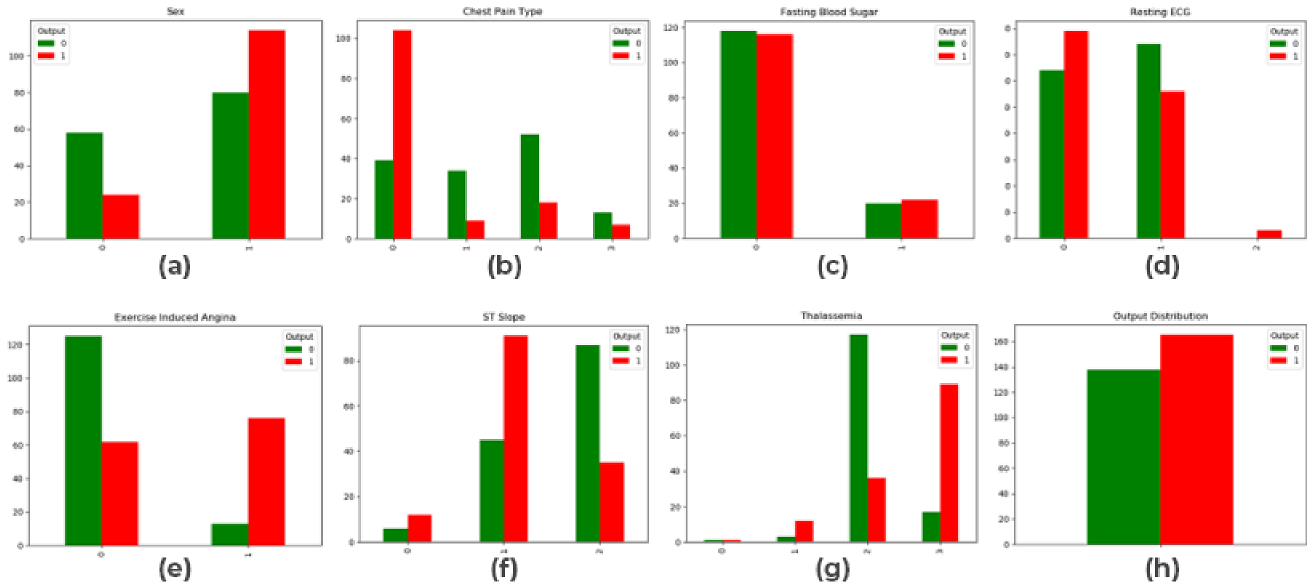


Figure 3. Histogram for the Categorical Attributes of the Dataset: (a) Sex, (b) Chest Pain Type, (c) Fasting Blood Sugar, (d) Resting ECG, (e) Exercise Induced Angina, (f) ST Slope, (g) Thalassemia, and (h) Output Distribution.

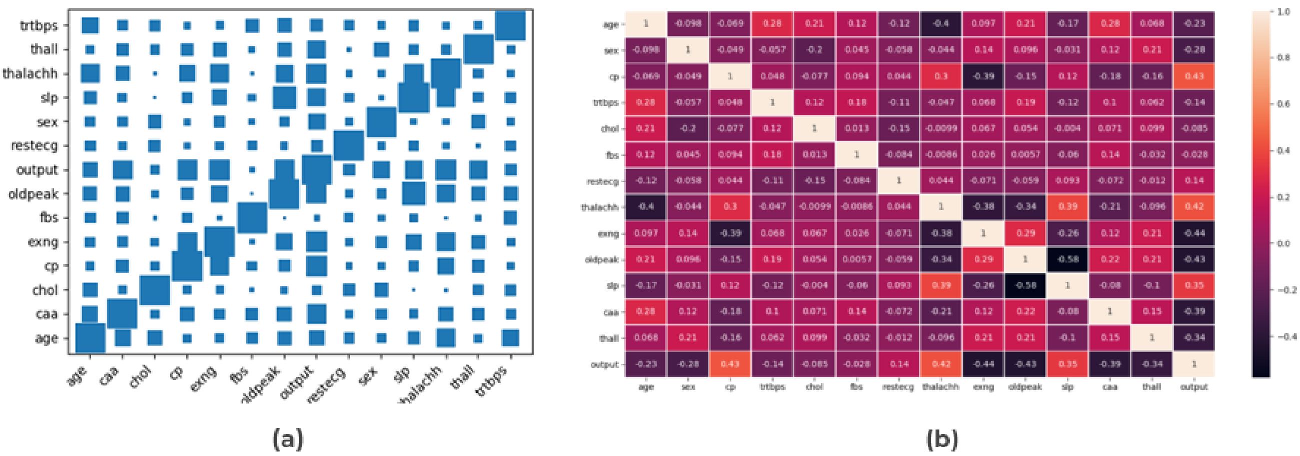
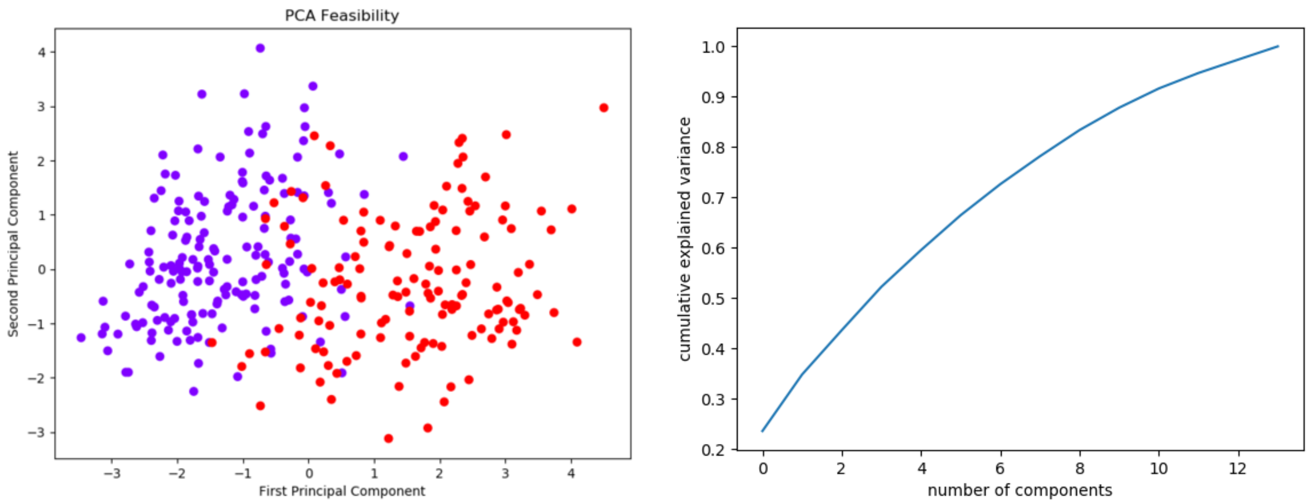


Figure 4. (a) Attribute-to-Attribute Co-variance Size Scatterplot and (b) Co-variance Heatmap.

Attribute Selection

The key features and the need to adjust the data were determined using Principal Component Analysis (PCA).

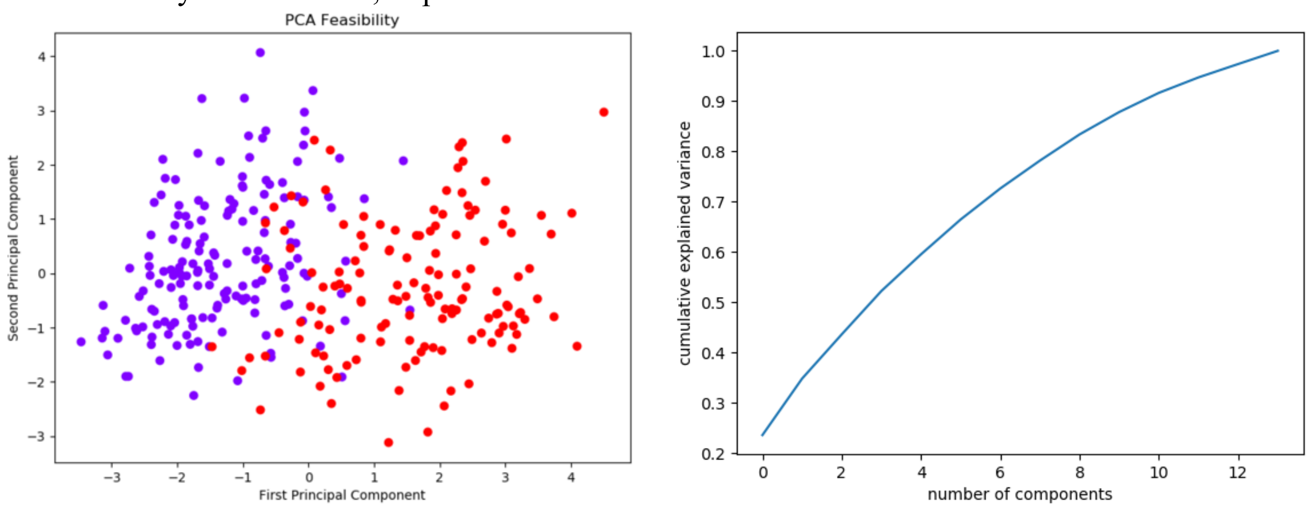
$$cov(A,\ B)\ =\ \frac{1}{n-1}\sum_{i=1}^{n}(B_i\ -\ \overline{B})(A_i\ -\ \overline{A})$$



(a) (b)

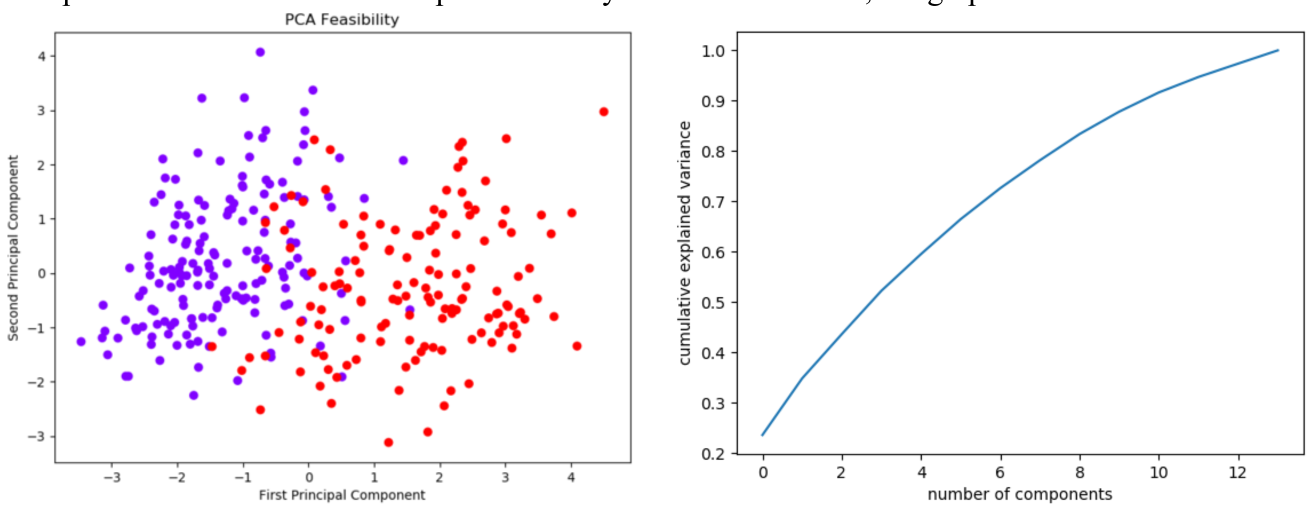
Figure 5. (a) PCA scatterplot for major two components showing glaring distinctive clusters and (b) PCA cumulative variance distribution for the 13 attributes.

The feasibility of the dataset, as presented in



(a) (b)

Figure 5a, is checked via a PCA cluster graph, which justifies the purpose of the dependency of independent constraints on the dependent entity. On the other hand, the graph in



(a) (b)

Figure 5b suggests the subtle variance, which is observed using every attribute, making it necessary to include every dependent quantity to predict the output and thus rules out the need for alteration of the Dataset.

Pre-Processing

The given Dataset is unbalanced and can lead to an imbalance of classes in training and exacerbate the accuracy and biased models, which justifies the reason for balancing the classes done by resampling the minority upscale it to have the same frequency as of majority class. Furthermore, the normalization of the numeric data is applied for simplifying the inputs, which facilitates the processing of training and testing of the Dataset. Class Balancing using resampling is done. Initially, the Dataset comprised 165 entries with '0' as output and 138 entries with '1' as output. Furthermore, it was upscaled to 330 entries with 165 from each of the classes to optimize the results.

Standardization

The process of tuning and rescaling characteristics so that the resultant attributes have a mean of 0 and a standard deviation of 1 is known as data standardization. It secures that the data is consistent. The following formula is used to calculate the standardized score.

$$stdizedValue = \frac{(x - \overline{x})}{\sigma}$$

Here, $\overline{x}$ is denoted as average, while σ depicts the standard deviation of the characteristic rows.

Train-Test Split

The Dataset is split into two sections: a training dataset that the model uses to fit to the Dataset, and a testing dataset that confirms and supports the trained model's validity. The dataset split in this study is 1:3, which implies that the training dataset makes up 75% of the whole dataset and the testing dataset makes up 25% of the total dataset.

MODELS AND METHODOLOGY INVESTIGATIONS

Random Forest

Decision Forests or Random Forests is an ensemble learning approach for regression as well as classification problems which involves training many individual decision tree

The Gini of each branch on a node, which specifies the more probable branches, is calculated using the class and probability formula. Here p_a denotes the occurrence, while *C* depicts the number of categories.

$$Gini = 1 - \sum_{a=1}^{C} (p_a)^2$$

Entropy is used to determine whether a node can branch depending on the likelihood of a specific outcome. Because it is derived using a logarithmic equation (Singh et al., 2017), it is more mathematically demanding than the Gini index.

$$Entropy = \sum_{i=1}^{(C} -\mathrm{pi} * \log_2(pi)$$

Supervised Learning Models

Supervised learning is a category of machine learning that uses named datasets to train algorithms to identify data or predict outcomes correctly. When input data is fed into the algorithm, it changes its weights using a reinforcement learning mechanism, ensuring that the model is adequately equipped.

There are 6 Supervised Learning Models which are utilized to compare and analyze the behavior and adaptability to the given Dataset.

- K- Nearest Neighbours
- Long Short-Term Memory (LSTM)
- Artificial Neural Network (ANN)
- Naïve Bayes
- Logistic Regression
- Support Vector Machine

K-Nearest Neighbours

The k-nearest neighbors (KNN) algorithm is a simple, easy-to-use supervised machine learning algorithm for classification and regression problems. The KNN algorithm assumes that identical objects happen nearby. In other terms, related entities are close together (Desai et al., 2021).

Algorithm for KNN:

Step 1: Loading the input data.
Step 2: Set K to the number of neighbors
Step 3: For each data example, calculate the distance from the data between the question example and the present example. To an ordered set, add the distance and the example's index.
Step 4: Sort the organized set of distances and indices by distance from smallest to greatest.
Step 5: Choose the first K entries from the sorted collection.
Step 6: Obtain the labels of the chosen K entries.
Step 7: If there is regression, return the average of the K symbols.
Step 8: If classification was performed, return the mode of the K marks.

Support Vector Machine

The Support Vector Machine (SVM) is a basic algorithm that any machine learning professional should have in his or her arsenal as the support vector machine. Many people choose support vector machines because they achieve substantial precision by using fewer computing resources. Hyperplanes are judgment boundaries that aid in the classification of data points. Different groups may be assigned to data points that land on either side of the hyperplane. Furthermore, the size of the hyperplane is determined by the number of functions. Support vectors are the data points that lie closer to the hyperplane that controls the direction and orientation of the hyperplane. Using these help vectors, we optimize the classifier's margin. The location of the hyperplane will change if the support vectors are removed. These are the points that will assist us in developing our SVM (Amami et al., 2015).

For a given Dataset $S = \{(x_1, t_1),\ (x_2, t_2), \ldots, (x_m, t_m) \mid (x_k, t_k)\ \in\ R^n * \{+1, -1\}\}$

Algorithm for SVM:

- Step 1: Create the template matrix.
- Step 2: Implement the Kernel Function and customize the parameters of Kernel Function and the value of C.
- Step 3: Run the training Algorithm to obtain the value of α
- Step 4: The value of α and support vectors are used to classify previously unseen data.

Mathematical Formula:

$$K(x_i, x_j) = x_i^T x_i$$

Logistic Regression

Logistic regression is a mathematical model which assesses if an independent variable influences a binary dependent variable. It implies that for a given input, there are only two possible outcomes.

A sigmoid function is a common model. The sigmoid function, also known as a squashing function, confines outputs to the range between 0 and 1.

$$y = \frac{e^{(b0+b1*x)}}{1 + e^{(b0+b1*x)}}$$

There are two variables, b0, and b1, in the function above. These are referred to as the weights or coefficient values. The bias, or intercept, is represented by b0, and b1 represents the coefficient. The actual data set is used to study and train these weights. This calculation would yield a percentage or expectation that will be mapped over discrete groups. The judgment boundary is the specified distinction between two groups (Manogaran & Lopez, 2018).

Naïve Bayes

Naïve Bayes classifiers are a subset of basic "probabilistic classifiers" based on applying Bayes' theorem to features with strict independence assumptions. They are among the most basic Bayesian network models, but they can reach higher levels of accuracy when combined with kernel density estimation.

To our Dataset, we can now apply Bayes' theorem as follows:

$$P(y|X) = \frac{P(X|y) * P(y)}{P(X)}$$

Where y is a class variable and X is an n-dimensional dependent function vector (Dulhare, 2018).

$$X = (x_1, x_2, x_3, \ldots, x_n)$$

ANN Sequential Model

Keras is an open-source deep learning platform for Python with a simple structure, which provides a clean and straightforward way to construct TensorFlow-based deep learning models. Keras Layers are the basic building blocks of Neural Networks, where a layer is made up of tensor-input and tensor-output calculation functions and a state in TensorFlow variables. Tensors are multidimensional arrays of a standardized data type. Tensors are both immutable.

Keras layer needs an input shape and an initializer to set the weight for each to understand the input data. The constraints parameter limits and defines the spectrum of input data weighting to be generated. The regularise optimizes the layer and the model by adding penalties to the weights dynamically during the optimization process. The input Shape must be given to the network's first layer.

Keras Layers are the rudimentary components of Neural Networks. A tensor-input and tensor-output computation function and a state stored in TensorFlow variables make up a layer. Tensors have several dimensions.

The Sequential model can be described as a network of Dense Layers utilized to train the deep learning framework. The proposed model uses Sequential Model, which enables a model fitted with multiple layers, where each layer has a weight that supplements the layer that follows it (Nalavade et al., 2014).

A total of 9 layers are applied, out of which one layer is for input, seven hidden layers, and the final layer is for the output purpose. All the layers except the last layer implement the Swish Activation, a soft curve and non-monotonic function and unbounded, which is a desirable attribute for an activation function. It can avoid possible problems when the gradients are nearly zero and thus outperform the traditional ReLU activation function. The swish function applies the following mathematical formula:

$$Y = X * \sigma(X)$$

While the final layer uses Sigmoid Activation to predict the binary class.

LSTM Sequential Model

The Long Short-Term Memory (LSTM) Network is an innovative RNN (sequential network) that allows information to be stored indefinitely. It will deal with the vanishing gradient problem that RNN has. For persistent memory, a recurrent neural network, also known as RNN, is used. RNNs recall facts from the past and apply them to the new input. Because of the vanishing gradient, RNNs are unable to remember long-term dependencies. Long-term dependence issues are expressly avoided for LSTMs (Islam et al., 2019). A total of three layers were associated with the Sequential Model to implement the LSTM functionalities, with the usage of 'Swish' activation.

Bagging Models

Bagging or bootstrap aggregating is a machine learning ensemble meta-algorithm for improving the consistency and accuracy of machine learning algorithms used for statistical classification and regression. It also helps to prevent overfitting by reducing variance. While it is most often associated with decision tree systems, it can be applied to any design. Bootstrap Aggregation, or bagging for short, is a straightforward and highly effective ensemble process.

The following models are ensembled with a bagging approach:

- Voting Classifier
- Extra-Trees
- Random Forest (Proposed Model)
- Decision Tree

Voting Classifier

The Voting classifier is not a classifier in and of itself but rather a wrapper for a set of different ones trained and evaluated in parallel to leverage the unique characteristics of each algorithm. Classifier, which involves voting of the other models to provide a unified class. Dataset can be trained using various algorithms and then use an ensemble to predict the final result (Latha & Jeeva, 2019). The Voting classifier has been utilized by implementing a selection of five KNN models with different neighbour values: KNN 1, KNN 3, KNN 5, KNN 7, and KNN 9.

Extra Trees

Extra Trees Classifier is an ensemble learning system focused on decision trees. Extra Trees Classifier, like Random Forest, randomizes some decisions and subsets of data to reduce over-learning and overfitting from data. It uses the same Formula as the Decision Tree. Extra Trees is like Random Forest. It creates several trees and divides nodes using random subsets of elements, but with two main differences: it does not bootstrap observations (samples without replacement). The nodes are divided into random splits rather than better splits. So, In all Extra Trees:

- Builds several bootstraps = False by example, so samples are divided into arbitrary divisions between a random subset of functions picked at each node without any substitute nodes.
- Nodes are divided based on arbitrary divisions between a random subset of selected features of each node.

Decision Tree

A decision tree is a decision-making method that employs a tree-like model of decisions and their potential outcomes, such as chance case outcomes, resource costs, and utility. It is one method of displaying an algorithm that consists solely of conditional control statements (Reddy et al., 2016).

Step 1: It starts with the initial root node, S.

Step 2: The algorithm iterates over the very unused attribute of the set S and measures the Entropy (H) and Information gain (IG) of this attribute for each iteration.

Step 3: It then chooses the attribute with the lowest Entropy or highest Information gain.

Step 4: The chosen attribute then subdivides the set S to yield a subset of the results.

Step 5: The algorithm proceeds to recurse on each subset, considering only attributes that have never been picked before.

$$\text{Entropy} = \sum_{i=1}^{(C} -\text{pi} * \log_2(pi)$$

$$\text{Gain}(\text{S}, \text{A}) = \text{Entropy}(\text{S}) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

Boosting Models

Boosting is an ensemble simulation technique that aims to create a robust classifier out of many weak ones. It is achieved by combining weak models to build a model. Firstly, a model is created using the training data. The second model is then developed, which attempts to correct the errors in the first model. This process is repeated until either the entire training data set is correctly estimated, or the total number of models has been applied. In all, six boosting models have been implemented on the Dataset to contrast the functionalities and results, namely:

- Gradient Boosting
- XGB
- CatBoost
- Gaussian Process
- LGBM
- AdaBoost
- SGD

Gradient Boosting

Gradient boosting classifiers are a set of machine learning algorithms that combine many poor learning models to construct a robust predictive model. When doing gradient boosting, decision trees are commonly used. The concept behind "gradient boosting" is to take a weak hypothesis or learning algorithm and make a set of tweaks to increase the hypothesis's power. The idea of Probability Approximately Correct (PAC) Learning underpins this method of hypothesis boosting (PAC).

$$\frac{\sum Residual}{\sum[PreviousProb * (1 - PreviousProb)]}$$

Sigma Residual is the sum of residuals of that specific leaf. The denominator suggests the sum of the prior probability of success and likelihood of failure (Bahad & Saxena, 2020).

XGB

XGB is an open-source software library that offers a regularising gradient boosting mechanism to provide Scalable, Portable, and Distributed Gradient Boosting. The following features of XGBoost distinguish it from other gradient boosting algorithms:

- Trees are penalized cleverly.
- A proportional reduction in the size of leaf nodes
- Boosting Newton
- An additional parameter for randomization
- Out-of-core computing and implementation on independent, distributed systems
- A Feature collection is made automatically.

XGBoost builds trees using the loss function, which minimizes the following value:

$$\mathrm{L}(\varphi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

$$\Omega(\mathrm{f}) = \Upsilon \mathrm{T} + \frac{1}{2}\lambda||w||^2$$

The goal is to find an optimal output value for the leaf to reduce the overall equation to a minimum. Since that starts with a value of y0, the following prediction is always equal to the i-1st forecast plus the output value from the ith tree. As a result, we replace the first element, as seen below.

$$L^{(t)} = = \sum_i l\left(y_i,\ \hat{y}_i^{(t-1)} +\ f_t(x_i)\right) + \Omega(\mathrm{f_t})$$

The Final Similarity score is calculated using the below formula [31].

$$L_{split} = \frac{1}{2}\left[\frac{(\sum_{i\in I_L} g_i)^2}{(\sum_{i\in I_L} h_i + \lambda)} + \frac{(\sum_{i\in I_R} g_i)^2}{(\sum_{i\in I_R} h_i + \lambda)} - \frac{(\sum_{i\in I} g_i)^2}{(\sum_{i\in I} h_i + \lambda)}\right] - \Upsilon$$

CatBoost

Yandex created CatBoost, an open-source software library. It offers a gradient boosting mechanism that, in contrast to the classical algorithm, aims to solve categorical features using a permutation-driven alternative. It provides out-of-the-box solid support for the more descriptive data formats that surround many market challenges. It produces state-of-the-art outcomes without the intensive data training that other machine learning approaches need. The procedure is as follows:

1. Randomly permute the set of input observations. Many random permutations are generated.
2. Converting a floating-point or unit mark value to an integer
3. The following function is used to convert all categorical attribute values to integer values:

$$\text{avg\_target} = \frac{\text{countInClass} + \text{prior}}{totalCount + 1}$$

CountInClass is the number of times the mark value for objects with the current categorical function value was equal to “1.”. The numerator’s preliminary value is called prior. The beginning parameters decide this. The cumulative number of items (up to the current one) with a categorical function attribute matching the current one is called TotalCount. It can be expressed mathematically using the following equation:

Let $\sigma = (\sigma_1, \ldots, \sigma_n)$ be the permutation, then $x_{\sigma_p,k}$ is substituted by

$$\frac{\sum_{j=1}^{p-1}\left[x_{\sigma_j,k} = x_{\sigma_p,k}\right] Y_{\sigma_j} + a.P}{\sum_{j=1}^{p-1}\left[x_{\sigma_j,k} = x_{\sigma_p,k}\right] + a}$$

Gaussian Process

The Gaussian Processes Classifier is a machine learning classification algorithm. Gaussian Processes generalizes the Gaussian probability distribution used to build advanced non-parametric machine learning algorithms for classification and regression. The distribution of random variables is summarised by Gaussian probability distribution functions, while the properties of the functions, such as the parameters of the functions, are summarised by Gaussian processes. As a result, Gaussian processes can be thought of as one degree of complexity or indirection above Gaussian functions. The concept behind Gaussian Process Regression is that we conclude that a series of observed values F_N at some points X_N correspond to the realization of a multivariate Gaussian Process with a prior distribution:

$$P(F_N | X_N) = \frac{e^{(-\frac{1}{2}F_N^T K_N^{-1} F_N)}}{\sqrt{(2\pi)^N \det(K_N)}}$$

The coefficients of K_N are expressed in terms of a correlation function (or kernel) $K_{mn} = K(x_m, x_n)$. The kernel's hyper-parameters are calibrated using the maximum probability theorem. K_N is chosen to represent a function's prior expectation, and thus the kernel's selection would have a substantial effect on the regression's correctness (Manogaran & Lopez, 2018)**.**

LGBM

LightGBM (Light Gradient Boosting Machine) is a free and open source distributed gradient boosting platform for machine learning created by Microsoft. It is used for ranking, sorting, and other machine learning tasks built on decision tree algorithms. Like CatBoost, it can accommodate categorical attributes by using function names as feedback. It is also simpler than one-hot coding and does not translate to one-hot coding. To determine the split worth of categorical functions, LGBM employs a unique algorithm. It employs two innovative techniques: Gradient-based One Side Sampling and Exclusive Feature Bundling (EFB), which address the drawbacks of the histogram-based algorithm found in most GBDT (Gradient Boosting Decision Tree) frameworks. The characteristics of the LightGBM Algorithm are formed by the two techniques of GOSS and EFB mentioned below. They work together to make the model run smoothly and give it an advantage over other GBDT systems. In the computation of knowledge gain, different data instances play different roles. The knowledge advantage would be more significant for samples with higher gradients.

AdaBoost

Adaptive Boosting is a predictive grouping meta-algorithm that can increase efficiency when combined with various learning algorithms. The results of the other learning algorithms are compiled into a weighted sum that represents the boosted classifier's final output. AdaBoost is adaptive in that it tweaks subsequent vulnerable learners in Favor of instances misclassified by prior classifiers. It could be less prone to the overfitting dilemma than other learning algorithms in certain situations. Individual learners will be poor, but if their success is marginally higher than random guessing, the final model may converge to a good learner. AdaBoost is a specific teaching system for boosted classifiers.

$$F_t(x) = \sum_{t=1}^{T} f_t(x)$$

Every ft is a poor learner that takes an object x as input and returns a value that indicates the object's class. The sign of the weak learner performance, for example, identifies the expected object type in the two-class puzzle, while the absolute value indicates the belief in that classification. Similarly, if the sample belongs to a positive class, the T^{th} classifier is positive; otherwise, it is negative (Fitriyani et al., 2020). For each example in the training set, each weak learner generates an output hypothesis, h(xi). A bad learner is chosen and given a coefficient α_t at each iteration *t* such that the sum training error *Et* of the resulting t-stage boost classifier is minimized.

$$E_t = \sum_i E[F_{t-1}(x_i) + \alpha_t h(x_i)]$$

$F_{t-1}(x)$ is the boosted classifier built up to the previous stage of the study, E(F) is any error feature, and $f_t(x) = \alpha_t h(x)$ is the slow learner under consideration for inclusion in the final classifier. Every iteration of the training method assigns a weight $w_{i,t}$ to each sample in the training set equal to the current error $E(F_{t-1}(x_i))$ on that sample. These weights can be used to inform the teaching of the slow learner; for example, decision trees that support separating sets of samples with high weights can be grown (Bahad & Saxena, 2020).

SGD

The term "stochastic" refers to a system or process that is subject to random chance. As a result, instead of selecting the entire data set for each iteration in Stochastic Gradient Descent, a few samples are chosen at random. The term "batch" is used in Gradient Descent to refer to the total number of samples from a dataset used to calculate the gradient for each iteration. The batch is taken to be the entire Dataset in traditional Gradient Descent optimization, such as Batch Gradient Descent. While using the whole Dataset is extremely useful for obtaining the minima with less noise, randomly, the problem arises when our datasets become large. This problem is solved using Stochastic Gradient Descent. SGD performs each iteration with a single sample, i.e., a batch size of one. The sample is randomly shuffled and chosen for iteration.

$$for\ i\ in\ range\ (m):$$

$$\theta_j = \theta_j - \alpha(\hat{y}^i - y^i)x_j^i$$

The gradient of the cost function of a single example is determined at each iteration rather than the sum of the slopes of all the samples. Since only one sample from the Dataset is selected at random for each iteration in SGD, the direction taken by the algorithm to reach the minima is generally noisier than in a traditional Gradient Descent algorithm. However, that does not matter because the path taken by the algorithm is irrelevant as long as we meet the minima in a slightly shorter amount of time.

EXPERIMENTAL RESULTS AND DISCUSSIONS

The Train-Test split approach splits the data arrays into two subsets, one for testing and the other for preparation. The Dataset has been divided into two parts: a 25% testing set and a 75% training set. The Random Forest algorithm has been applied to predict the dataset with utmost accuracy. The proposed random forest framework is measured in terms of accuracy for metrics, and scores are compared based on sensitivity, specificity, accuracy, precision, and F1-score. All the scores are derived from the confusion matrix. For the binary classification, the model predicts the output '1', when the person is likely to suffer from heart disease, and '0' if the person is not expected to suffer from heart disease. For every classified output, it can be categorized into four categories as presented in Figure 6:

- **True Positive**: when the person has suffered from heart disease, and the model predicts the same.
- **True Negative**: when the person has not suffered from heart disease and the model predicts the same.
- **False Negative**: when the person has suffered from heart disease, and the model predicts the opposite.
- **False Positive**: when the person has not suffered from heart disease, and the model predicts the opposite.

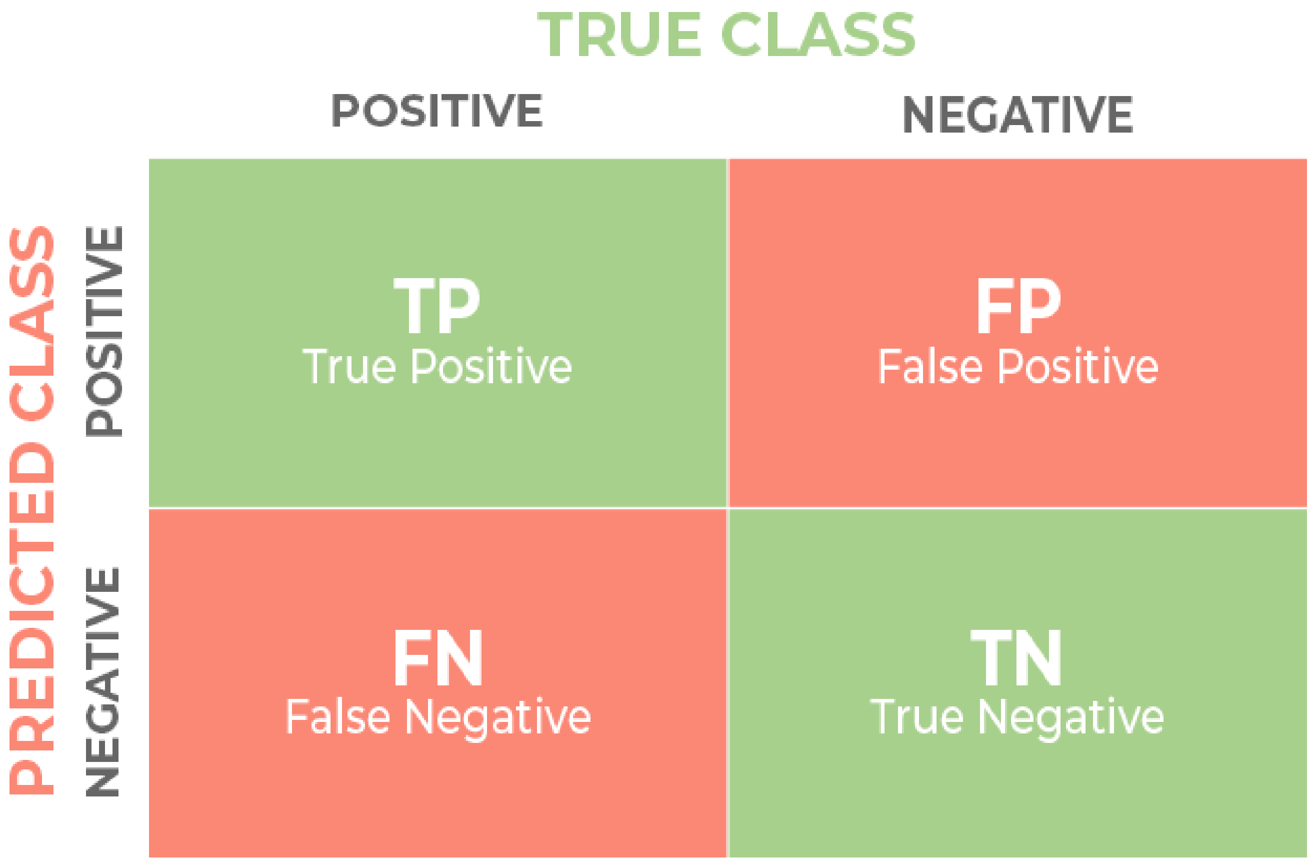


Figure 6. Confusion Matrix Category Distribution

Table 3. Parameters for the Bagging Models

Model Type	**n_estimators**	**Voting**	**max_features**
Voting Classifier	KNN1, KNN3, KNN5, KNN7, KNN9	Soft	-
ExtraTrees Classifier	100	-	sqrt
Random Forest	300	-	sqrt
Decision Tree	-	-	sqrt

Table 4. Parameters for the Supervised Learning Models

Model Type	**Parameters**
SVM	C: [0.1, 1, 10, 100, 1000], Gamma = [1, 0.1, 0.01, 0.001, 0.0001]
KNN	K value: 49
Logistic Regression	-
Naive Bayes	-
ANN	Dense Layers: 12/32/128/256/256/64/16/8/1, Activation: Swish, Sigmoid for the last layer

LSTM	LSTM Layers: 64/512/2048/128, Activation: Swish, Sigmoid for the last layer

Table 5. Parameters for the Boosting Models

Model Type	Parameters
SGD	max_iter = 1000, tol = 0.01
AdaBoost	n_estimators = 50, learning_rate = 1
LGBM	-
Gaussian Process	kernel = 1.0 * RBF (1.0)
CatBoost	n_estimators = 100
XGB	n_estimators = 10, objective ='reg:linear', colsample_bytree = 0.3, learning_rate = 0.1, max_depth = 5, alpha = 10
Gradient Boosting	-

Table 6. Analysis of Scores of Bagging Models

Model Type	Recall	F1 Score	Precision	Specificity	Sensitivity	Testing Accuracy	Training Accuracy
Decision Tree	86	100	91	91	86	88.41	100
Random Forest	92	100	100	100	92	95.65	100
Extra Trees	94	100	94	100	89	94.2	100
Voting Classifier	75	100	75	76	75	75.36	90.34

Table 7. Analysis of Different Supervised Learning Models

Model Type	Recall	F1 Score	Precision	Specificity	Sensitivity	Testing Accuracy	Training Accuracy
SVM	83	100	83	88	78	82.61	67.63
KNN	78	100	90	91	78	84.06	63.77
Logistic Regression	92	100	92	91	92	91.3	84.54
Naive Bayes	88	100	88	91	86	88.41	82.61
ANN	92	100	97	97	92	94.2	100
LSTM	92	100	97	97	92	94.16	100

Table 8. Analysis of Scores of Boosting Models

Model Type	Recall	F1 Score	Precision	Specificity	Sensitivity	Testing Accuracy	Training Accuracy
SGD	80	100	80	94	67	79.71	68.6
AdaBoost	88	100	88	88	89	88.41	95.65
LGBM	93	100	93	97	89	92.75	100
Gaussian Process	91	100	91	97	86	91.3	87.92
CatBoost	93	100	93	97	89	92.75	92.27
XGB	86	100	86	94	78	85.51	84.54
Gradient Boosting	89	100	89	95	84	89.16	100

The performance of the state-of-the-art machine learning approaches like Gradient Boosting Classifier, SGD Classifier, Cat Boost Classifier, LGBM Classifier, XGB Classifier, AdaBoost Classifier, Gaussian Process Classifier, LSTM RNN Model, Keras Sequential ANN Model, Naïve Bayes, Logistic Regression, KNN, Support Vector Machine, Voting Classifier, Decision Tree, and ExtraTrees Classifier is juxtaposed with the Random Forest model, also depicted in Tables 4 to 8. According to the results, the Random Forest model produces optimal efficiency. Figures 7 to 9 depict the comparative performance assessments of the classifier models by considering all popular classification algorithms to identify the higher accuracy classifier model to predict heart disease automatically. When it comes to accuracy, we discovered that a less computationally intensive strategy like random forests outperforms a more complicated technique like Neural Networks. The key reason for this is the tiny size of our Dataset, which can forecast the likelihood of cardiac arrest. The experiments were run with and without feature selection to assess the impact of feature selection on heart attack prediction.

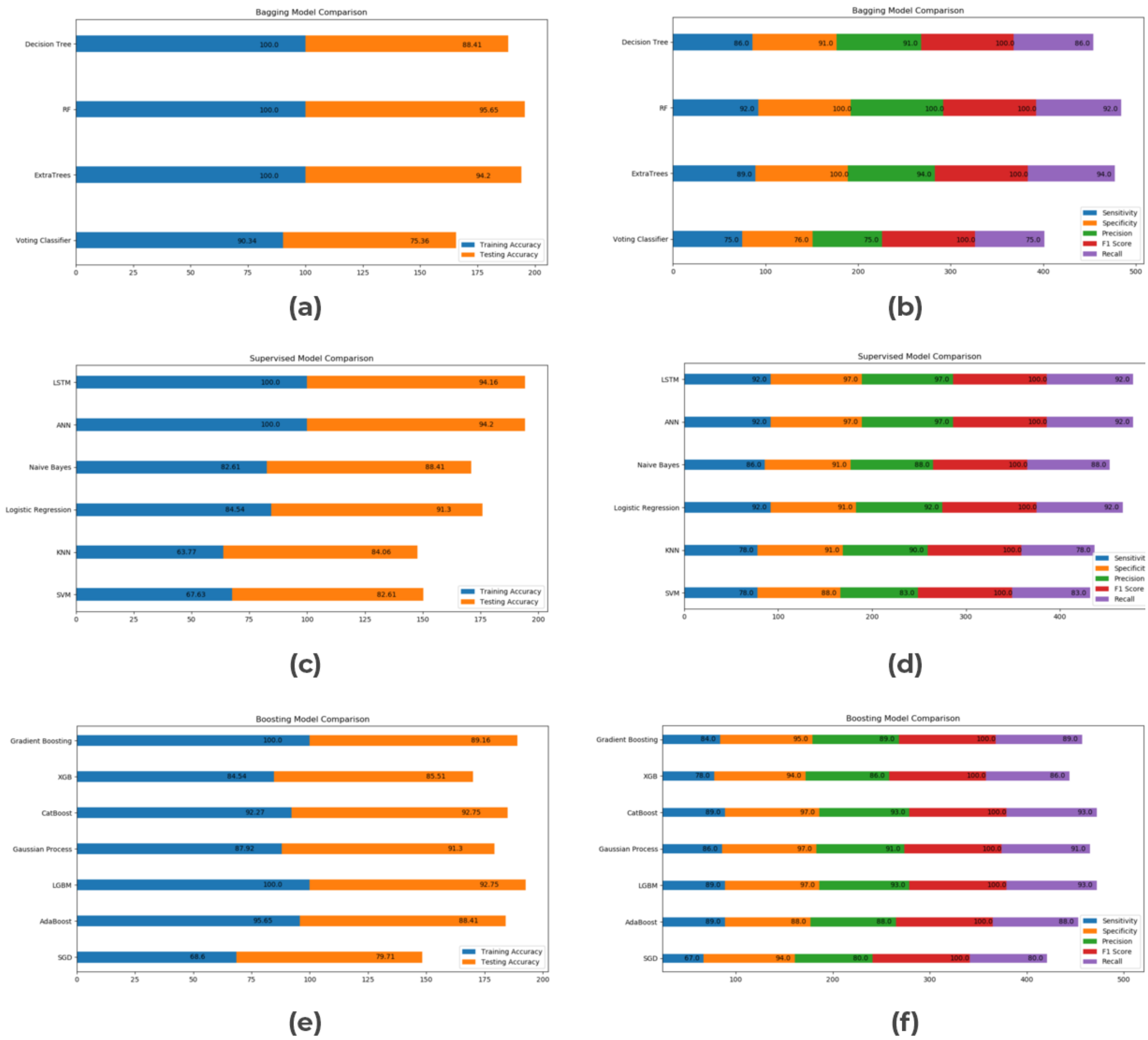


Figure 7. Training and Testing Accuracy of (a) Bagging, (c) Supervised, and (e) Boosting Models and the Recall, F1-Score, Precision, Specificity, and Sensitivity (b) Bagging, (d) Supervised, and (f) Boosting Models.

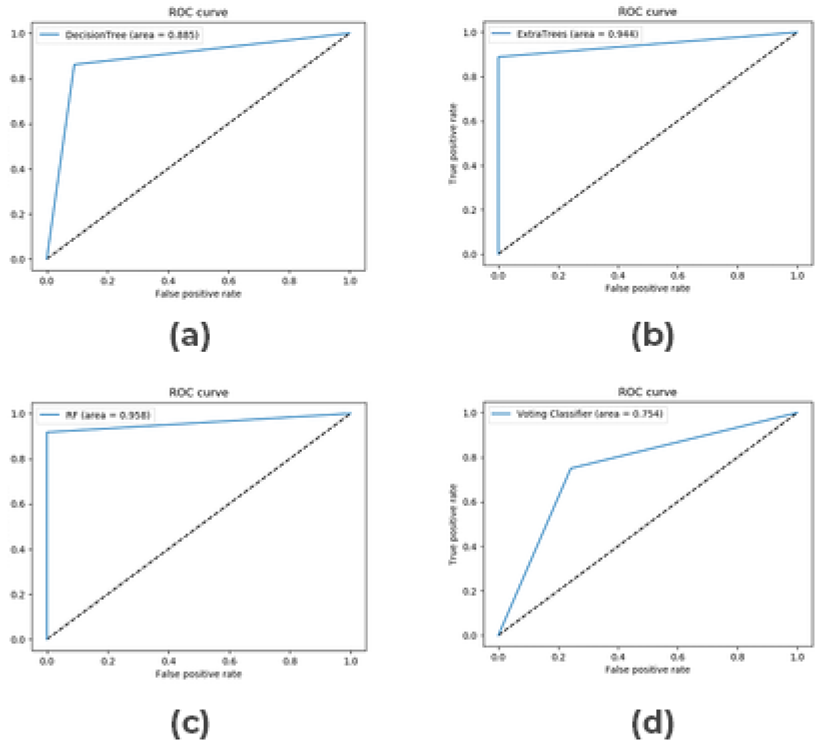


Figure 8. ROC Curve for Bagging Models: (a) Decision Tree, (b) Extratrees, (c) Random Forest, and (d) Voting Classifier

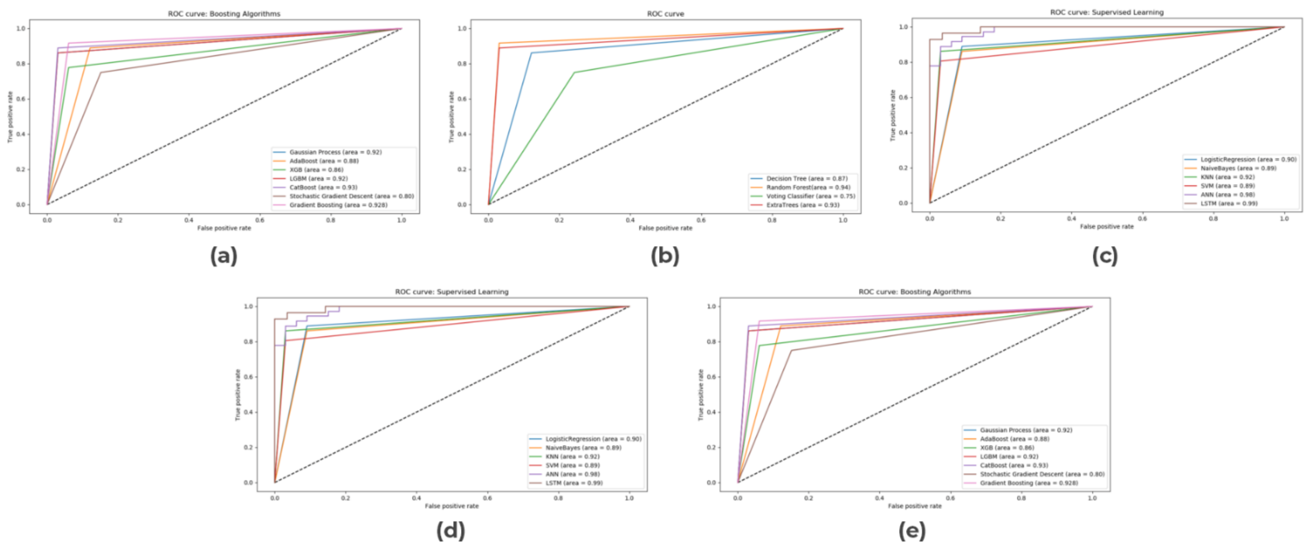


Figure 9. ROC Curve for (a) Boosting, (b) Bagging, and (c) Supervised Models and Compiled ROC Curves for Supervised and Boosting Models.

SOCIAL HEALTH EDUCATIONAL IMPLICATIONS

A visual analysis may accurately predict cardiac arrest, making it a potent educational tool for raising public awareness of health issues. By predicting cardiac arrest earlier, preventative steps can be taken to save lives. Promoting such health knowledge in society can dramatically lower the world mortality rate. Cardiovascular disease is a term used to describe conditions affecting the heart and blood arteries. Heart disease-related deaths have been rising quickly. The deadliest cause of death worldwide is thought to be cardiovascular disease. The heart disease dataset from the UCI Machine Repository is used in this study to provide an accurate diagnosis of heart illness. Using an optimization algorithm to diagnose heart disease can be beneficial in terms of improved sensitivity and accuracy. Optimization is the process of selecting the optimal solution from all feasible solutions to a given problem. Through the prediction of intelligent cardiovascular disease, the best outcomes are produced by under-researched categorization algorithms when they are equipped with a gradient descent optimization model. The need to create a model for accurately and efficiently diagnosing heart disease has increased due to the rise in heart failure-related fatalities. The provision of dependable facilities at reasonable prices is a major issue for healthcare institutions (healthcare centres, hospital clinics). The key issues with the current models are accuracy, utility, and reliability. The goal of the study is to identify the most efficient machine-learning method for making more precise, sensitive, and accurate earlier predictions of heart disease so that curative actions can be taken in advance to save human lives in society.

CONCLUSION AND FUTURE WORKS

Visual analysis is a powerful learning tool for promoting health awareness in society by earlier predicting cardiac arrest. Visual analysis is a powerful instructional tool for educating the public about health issues since it may properly predict cardiac arrest. Preventative measures to save lives can be implemented via earlier cardiac arrest prediction. A significant reduction in global death rates can be achieved by promoting such health information in society. This research uses the Random Forest approach to predict heart disease automatically. The experiment was conducted using the publicly available heart disease dataset. Before utilizing the input data to train and test the suggested model, it was necessary to undertake class balance through up-sampling minority classes and data standardization using the Z-score approach. Seventeen eclectic machine-learning techniques have been applied for comparing the alignment of the proposed model. The random forest produces the highest efficiency, according to empirical evidence. In the future, the suggested approach will be evaluated against a fresh heart disease dataset and used to forecast heart disease on a web-based platform to enable global accessibility through low-end computing devices.

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KEY TERMS AND DEFINITIONS

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